
CSC 412 – Operating Systems

Lab Session 05 [Lab 04 Part II], Fall 2025

Tuesday, October 7th and Thursday, October 9th, 2025

What this Lab Session Is About

In this lab we are simply going to finish the job we started in Lab 04, and more specifically look at the `execXY()` family of functions.

1 The Handout Code

1.1 The partial solution

The partial solution folder contains solutions for Versions 1 and 2 (and also Version 3a) of Lab 04.

1.2 Make sure that you understand the code

Look at the code of the solutions to Versions 1 and 2. Build and execute them. Make sure that you understand them before we move to Version 3.

2 Version 3

In this version, the child processes are going to perform an `execXY()` to flush the code and data of the parent process and start running the code of one of the four image processing utilities. The basic idea is that the child process looks at the command string it got, and in particular at the first “word” in that string, which is the name of the executable to launch.

I remind you that there is no time for data validation in labs (unless some day we have a lab explicitly and exclusively about data validation). We have to assume that the input is valid, because there is not time to do checking. So here, the first work of a `TASK_LIST` entry can only be one of `flipV`, or `flipH`.

2.0.1 Version 3a

I provide you a solution for this version as well. This one constitutes an intermediate step toward the complete solution. For each of the four executables, I hard-code a valid `execXY` command, to make sure that you actually launch the proper executable and that the command’s arguments are passed along as you expect.

2.0.2 Version 3b

Now, you want to use the string you got from the `TASK_LIST` to produce a valid `execXY` command.